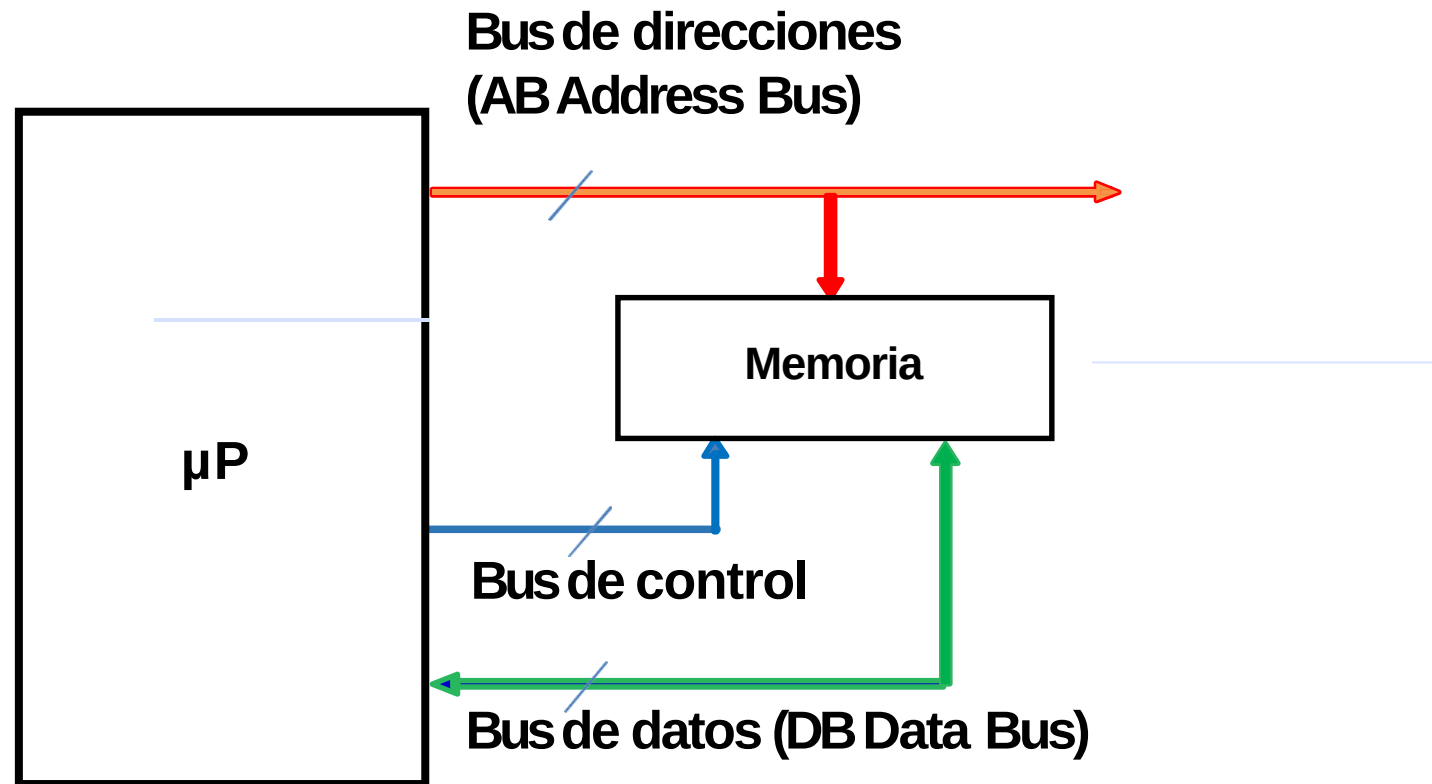
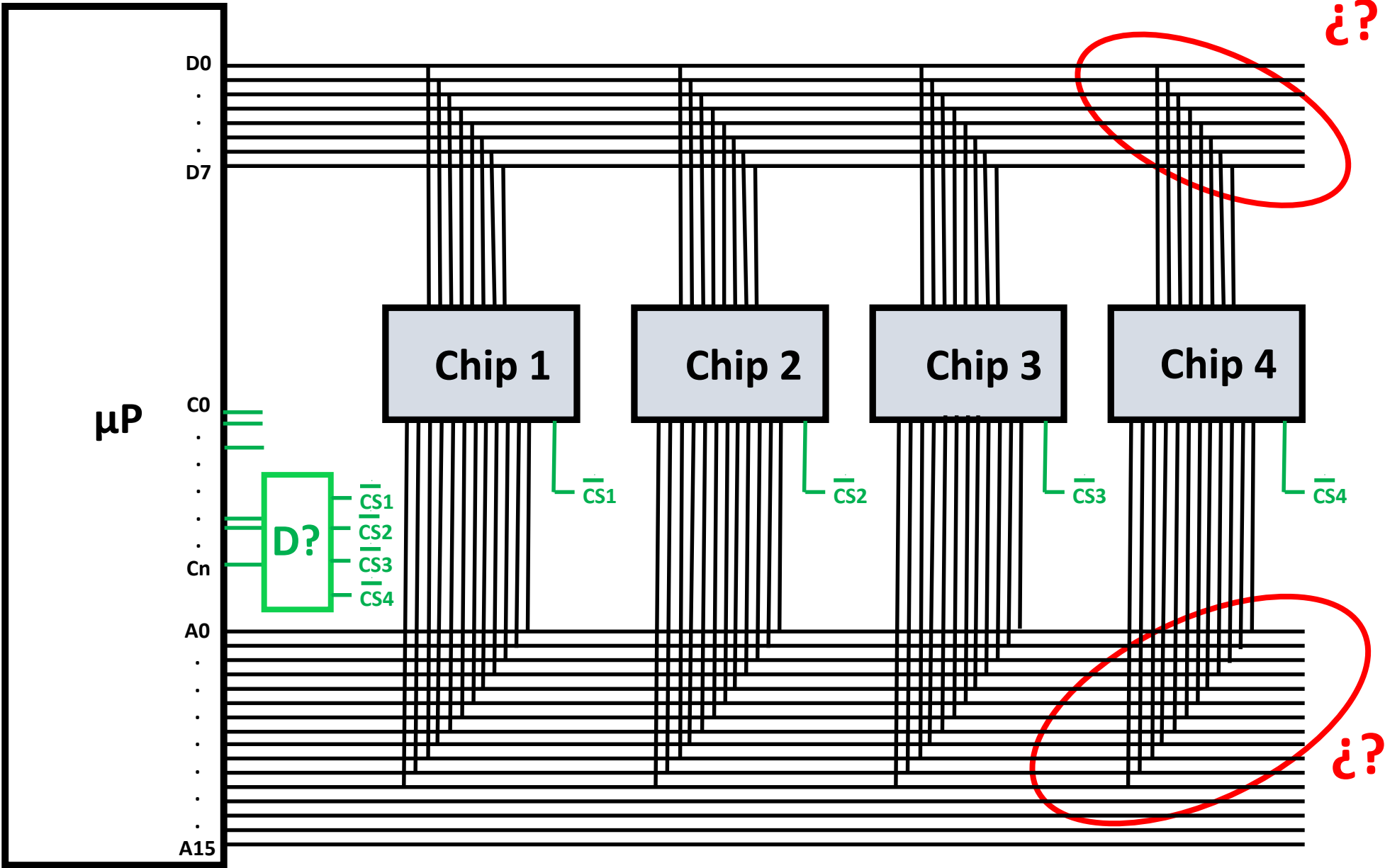


Mapeo de Memorias



Planteo del problema



Microprocesador $\Rightarrow$ 8 bits datos – 16 bits direcciones

16 bits direcciones:

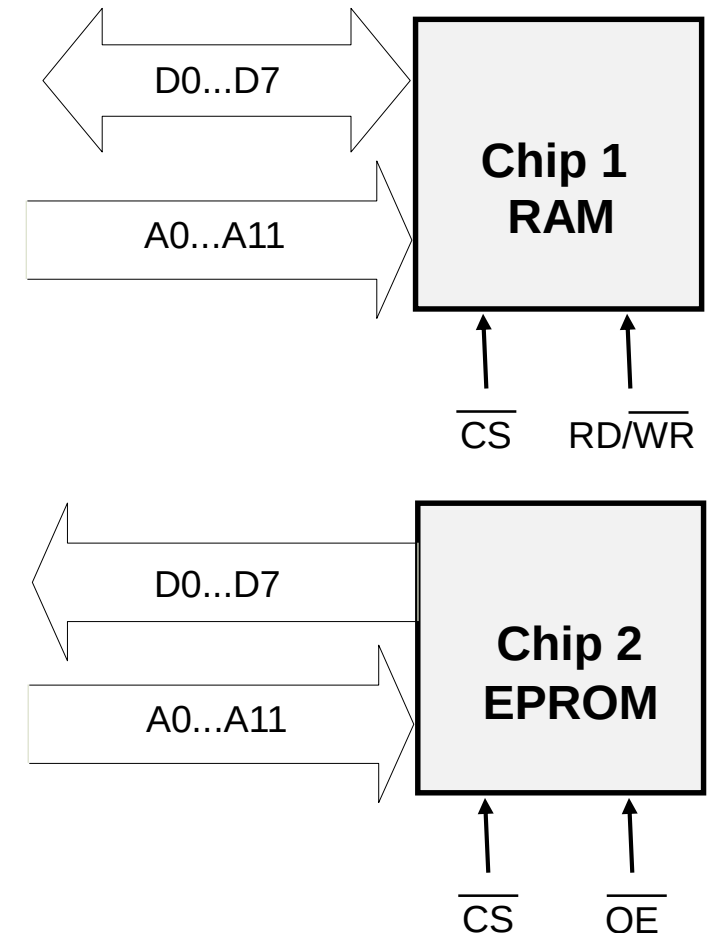
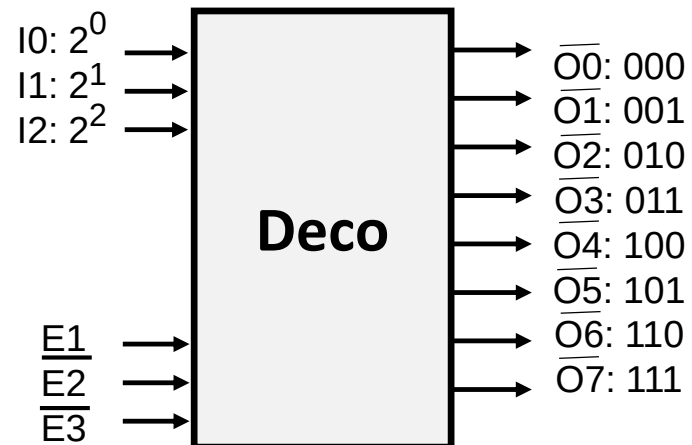
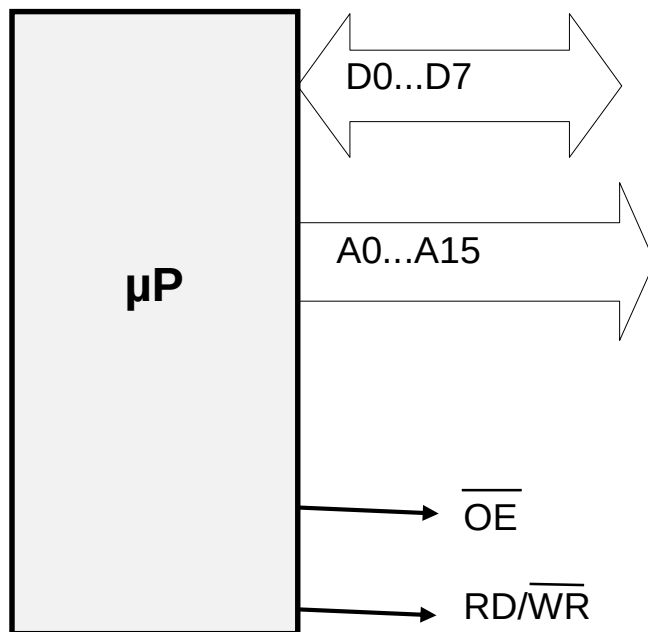
$\Rightarrow$ 0000 0000 0000 0000 0000b a 1111 1111 1111 1111b

$\Rightarrow$ 0000 a FFFF

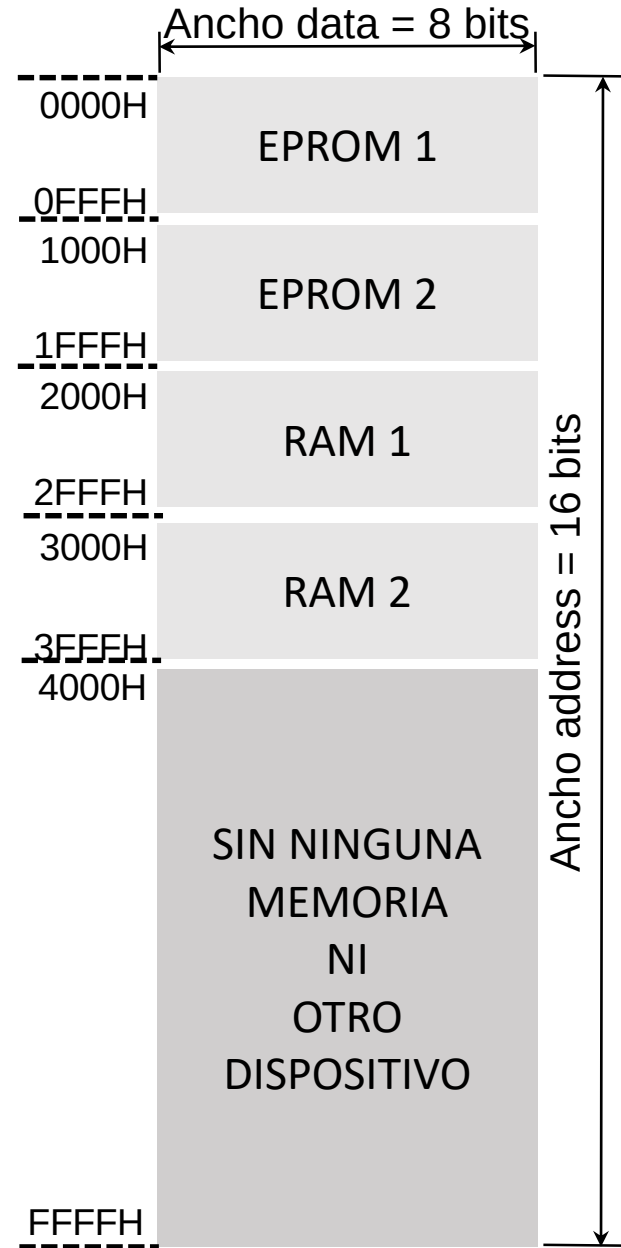
$\Rightarrow$ 0_{10} a $65535_{10} = 64 \text{ kB}$

Chips de memorias $\Rightarrow$ 4 k x 8 bits = 4kBytes

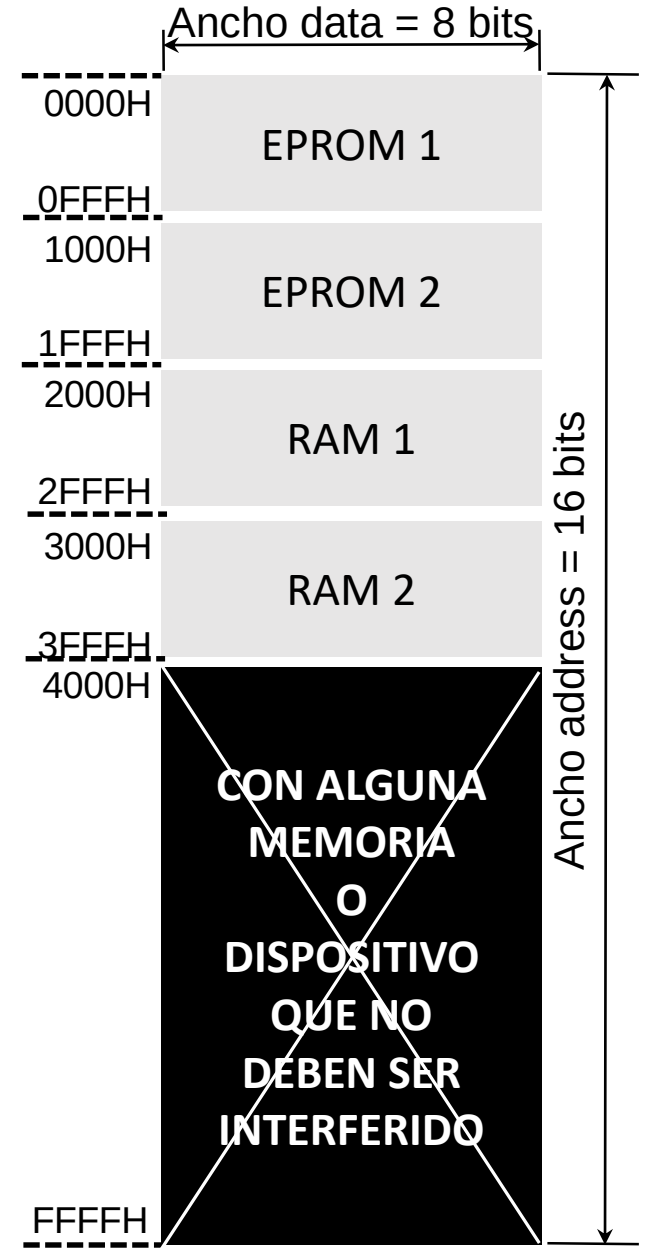
Decodificador $\Rightarrow$ Salida negada 1 en 8



Mapa de memoria



decodificación relativa



decodificación absoluta

Bits de selección						Bits de dirección							
A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	...	A ₀	Rango	Bytes	
0	0	0	0	0	0	0	0	0	...	0	0000 0FFF	4K	EPROM#1
0	0	0	0	1	1	1	1	1	...	1			
0	0	0	1	0	0	0	0	0	...	0	1000 1FFF	4K	EPROM#2
0	0	0	1	1	1	1	1	1	...	1			
0	0	1	0	0	0	0	0	0	...	0	2000 2FFF	4K	RAM#1
0	0	1	0	1	1	1	1	1	...	1			
0	0	1	1	0	0	0	0	0	...	0	3000 3FFF	4K	RAM#2
0	0	1	1	1	1	1	1	1	...	1			

Decodificación Relativa

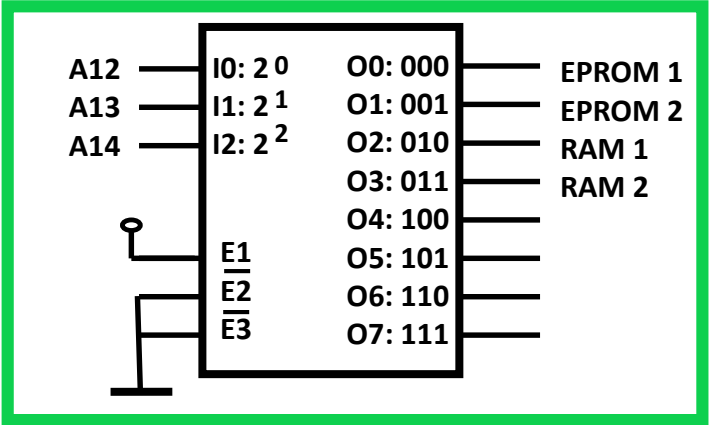
Bits de selección						Bits de dirección							
A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	...	A ₀	Rango	Bytes	
1	0	0	0	0	0	0	0	0	...	0	8000 8FFF	4K	EPROM#1
1	0	0	0	1	1	1	1	1	...	1			
1	0	0	1	0	0	0	0	0	...	0	9000 9FFF	4K	EPROM#2
1	0	0	1	1	1	1	1	1	...	1			
1	0	1	0	0	0	0	0	0	...	0	A000 AFFF	4K	RAM#1
1	0	1	0	1	1	1	1	1	...	1			
1	0	1	1	0	0	0	0	0	...	0	B000 BFFF	4K	RAM#2
1	0	1	1	1	1	1	1	1	...	1			

Técnica de resolución

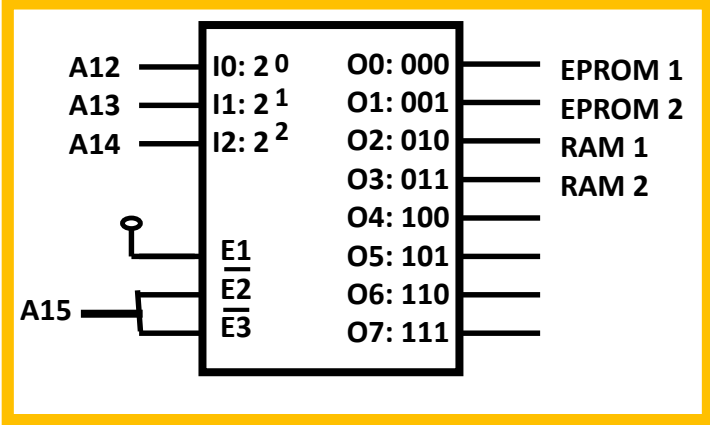
Bits de selección						Bits de dirección								
A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	...	A ₀	Rango	Bytes		
0	0	0	0	0	0	0	0	0	...	0	0000 0FFF	4K	EPROM#1	
0	0	0	0	1	1	1	1	1	...	1				
0	0	0	1	0	0	0	0	0	...	0	1000 1FFF	4K	EPROM#2	
0	0	0	1	1	1	1	1	1	...	1				
0	0	1	0	0	0	0	0	0	...	0	2000 2FFF	4K	RAM#1	
0	0	1	0	1	1	1	1	1	...	1				
0	0	1	1	0	0	0	0	0	...	0	3000 3FFF	4K	RAM#2	
0	0	1	1	1	1	1	1	1	...	1				

Decodificación:

Relativa



Absoluta



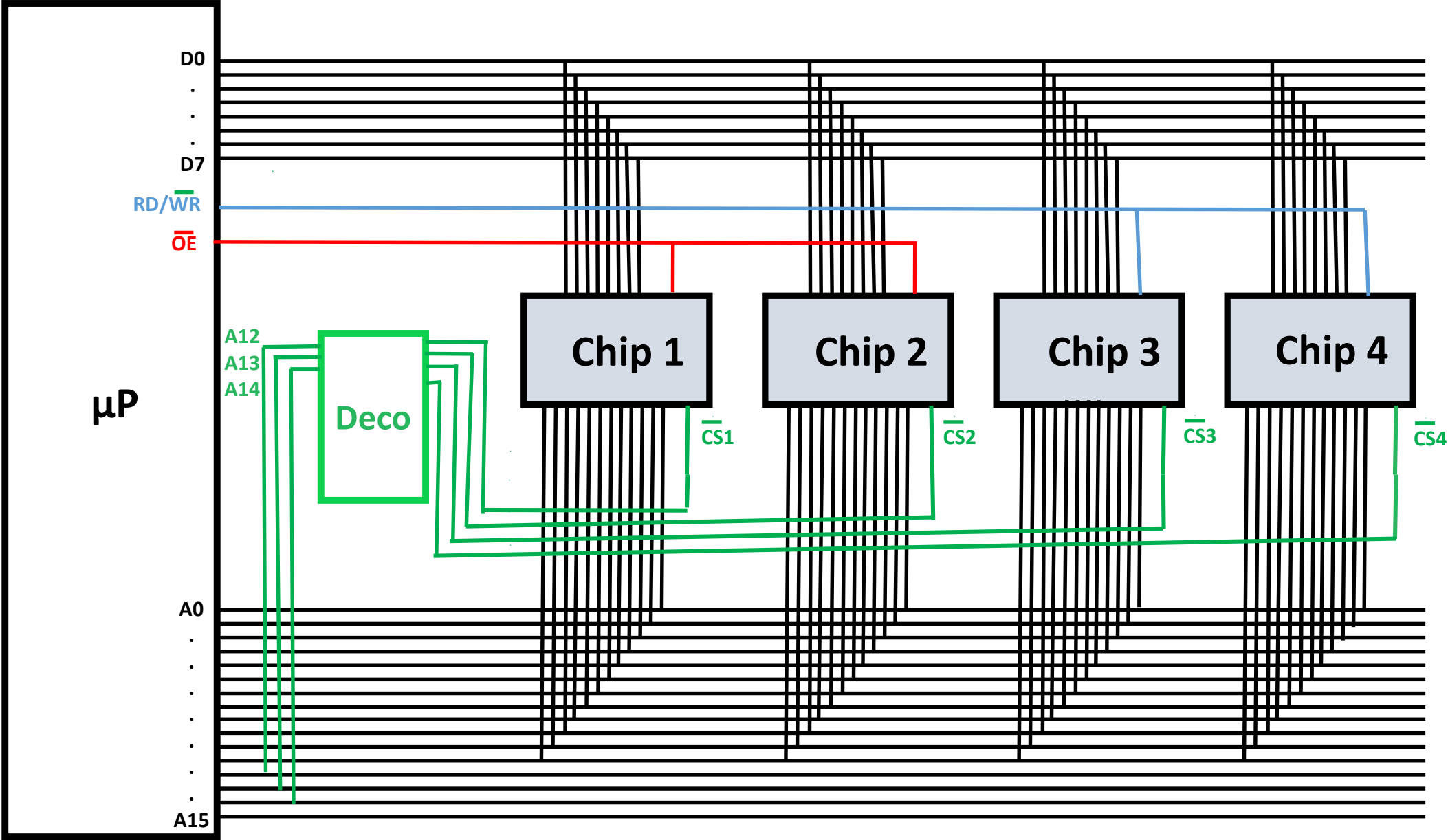
decodificación relativa

0000H - 0FFFH	EPROM 1
1000H - 1FFFH	EPROM 2
2000H - 2FFFH	RAM 1
3000H - 3FFFH	RAM 2
4000H - 7FFFH	NADA
8000H - 8FFFH	EPROM 1
9000H - 9FFFH	EPROM 2
A000H - AFFFH	RAM 1
B000H - BFFFH	RAM 2
C000H - FFFFH	NADA

decodificación absoluta

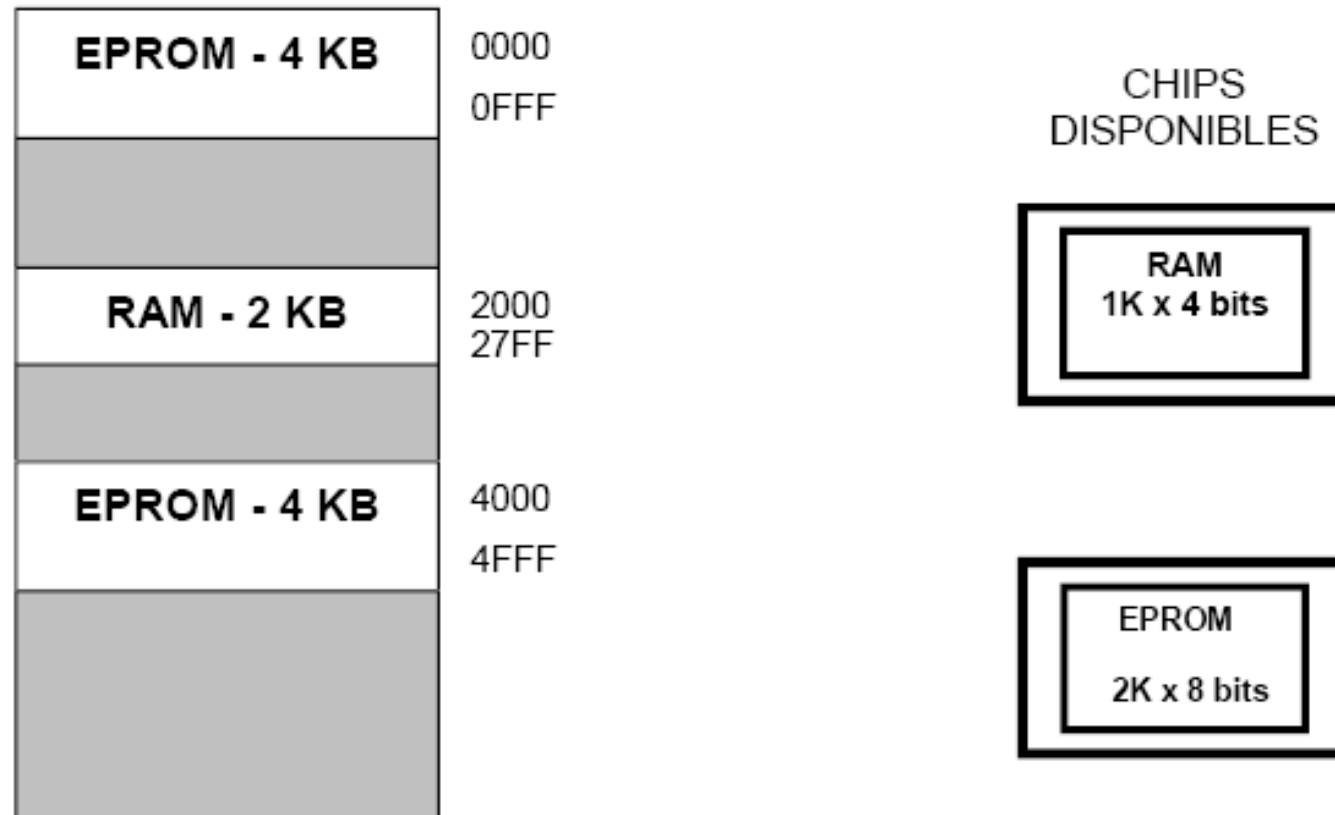
0000H - 0FFFH	EPROM 1
1000H - 1FFFH	EPROM 2
2000H - 2FFFH	RAM 1
3000H - 3FFFH	RAM 2
4000H - FFFFH	NADA

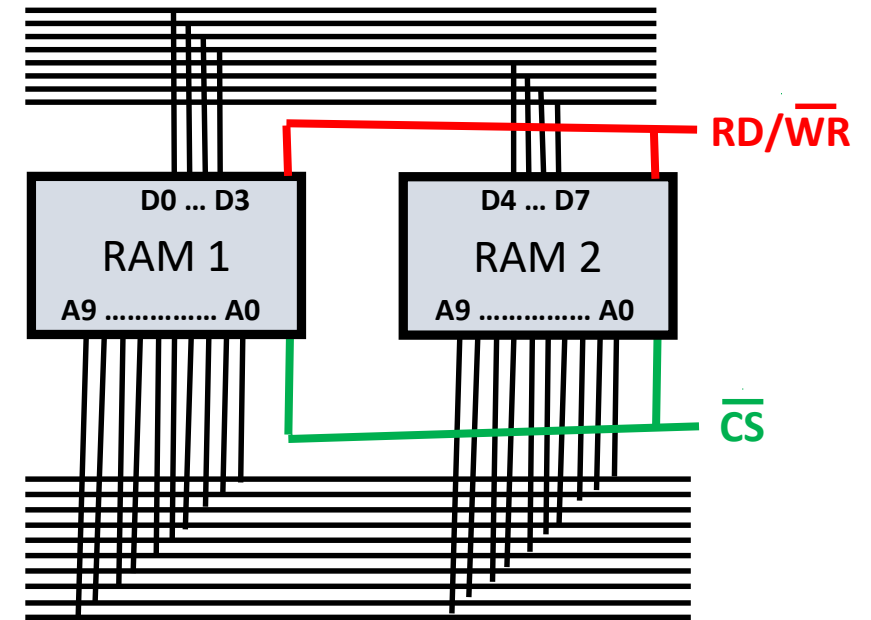
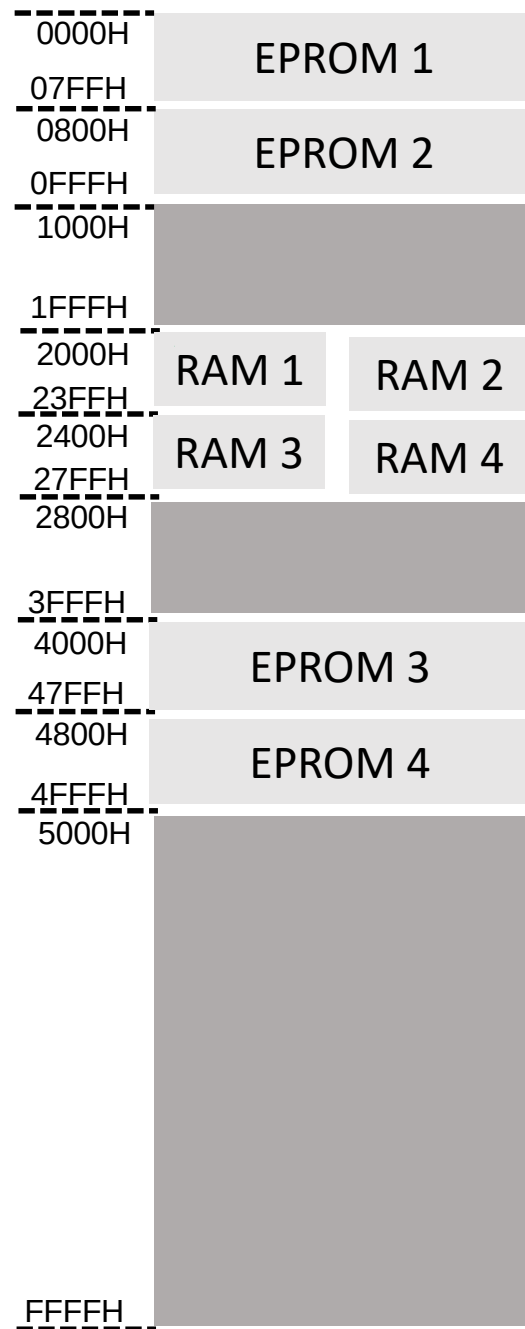
Circuito completo



Problema 1.1.

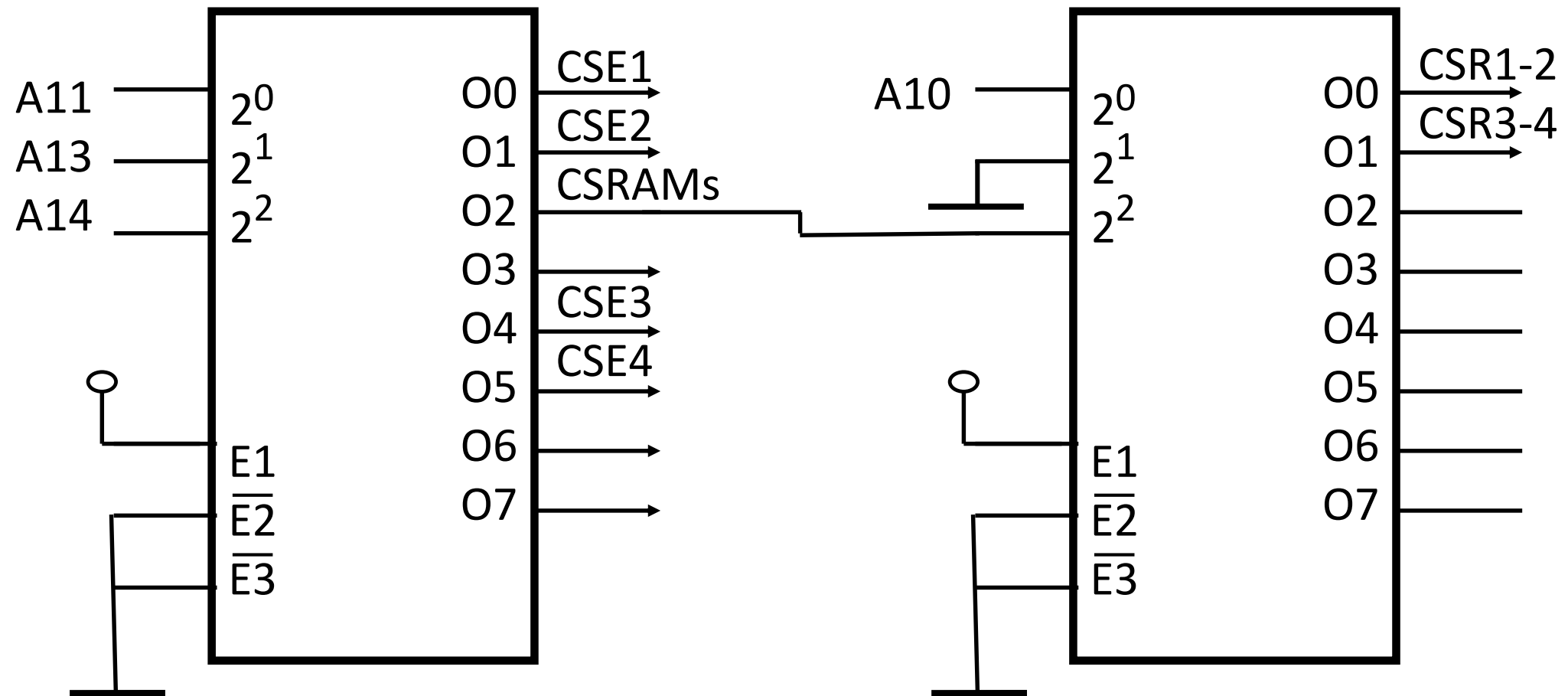
Implementar un Banco de Memoria que responda al siguiente mapa:

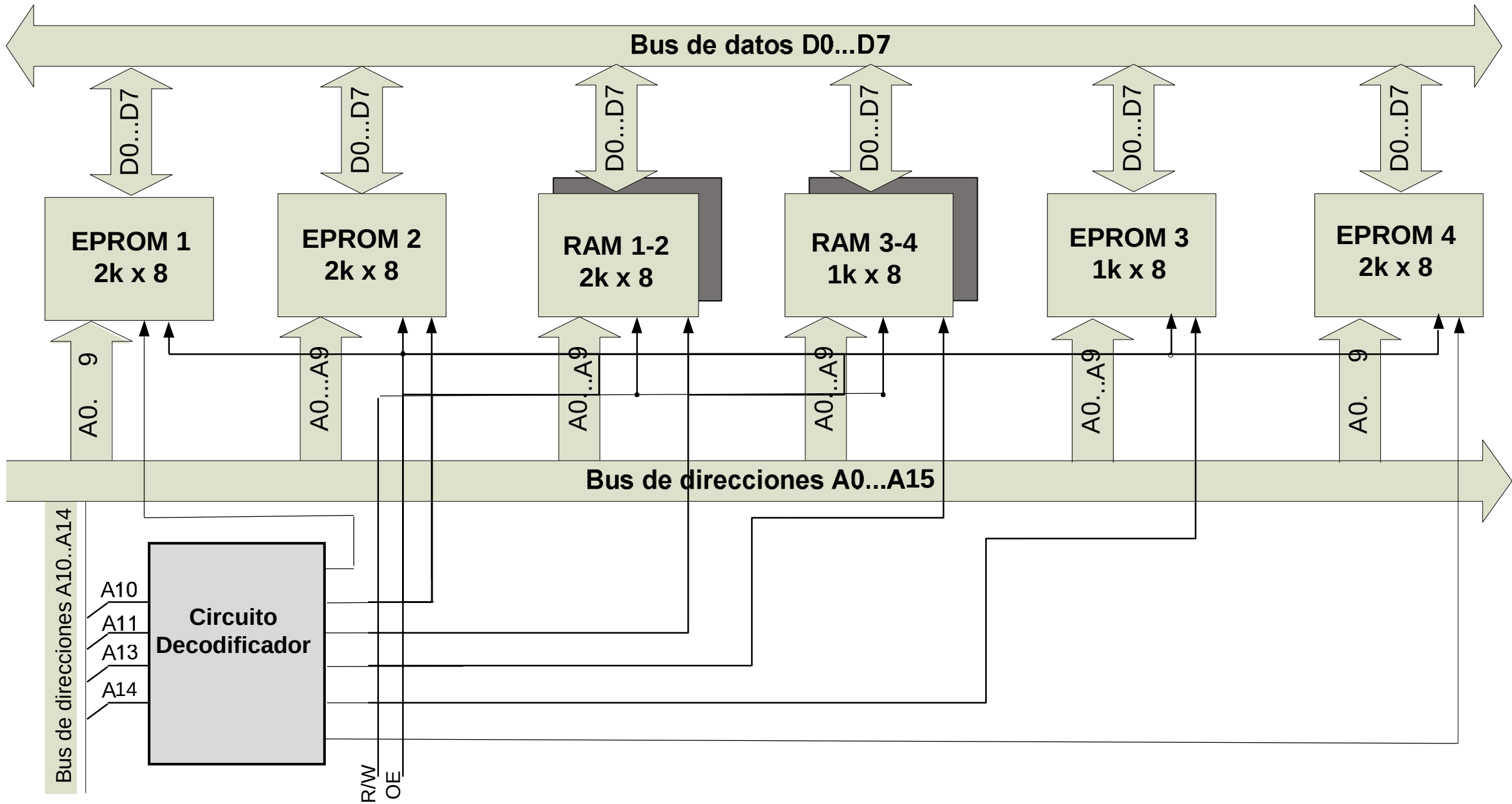




Bits de selección						Bits de dirección							
A ₁₅	A ₁₄	A ₁₃	A ₁₂	A ₁₁	A ₁₀	A ₉	A ₈	A ₇	...	A ₀	Rango	Bytes	
0	0	0	0	0	0	0	0	0	...	0	0000 07FF	2K	EPROM#1
0	0	0	0	0	1	1	1	1	...	1			
0	0	0	0	1	0	0	0	0	...	0	0800 0FFF	2K	EPROM#2
0	0	0	0	1	1	1	1	1	...	1			
0	0	1	0	0	0	0	0	0	...	0	2000 23FF	1K	RAM#1-2
0	0	1	0	0	0	1	1	1	...	1			
0	0	1	0	0	1	0	0	0	...	0	2400 27FF	1K	RAM#3-4
0	0	1	0	0	1	1	1	1	...	1			
0	1	0	0	0	0	0	0	0	...	0	4000 47FF	2K	EPROM#3
0	1	0	0	0	1	1	1	1	...	1			
0	1	0	0	1	0	0	0	0	...	0	4800 4FFF	2K	EPROM#4
0	1	0	0	1	1	1	1	1	...	1			

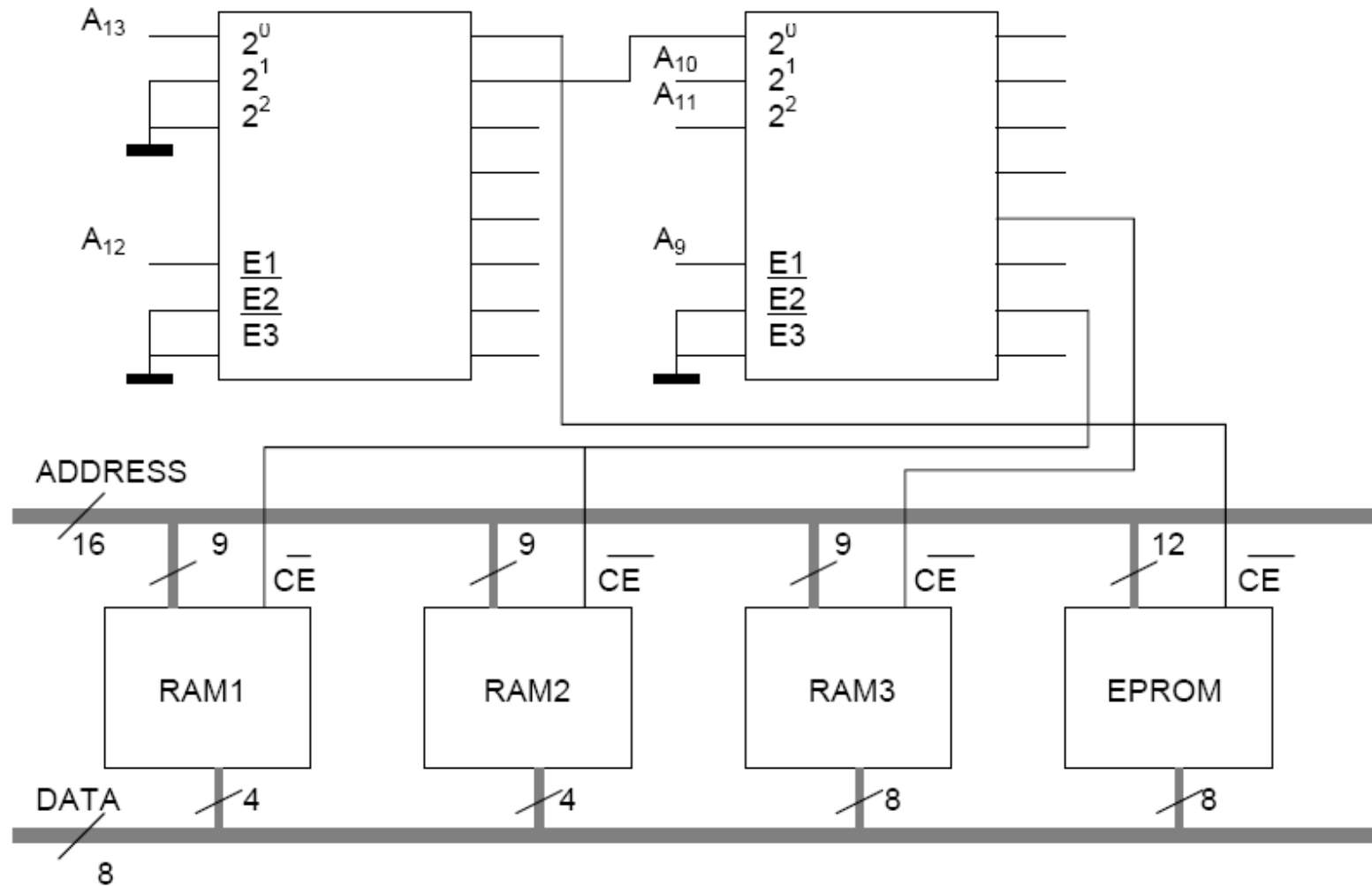
Decodificación

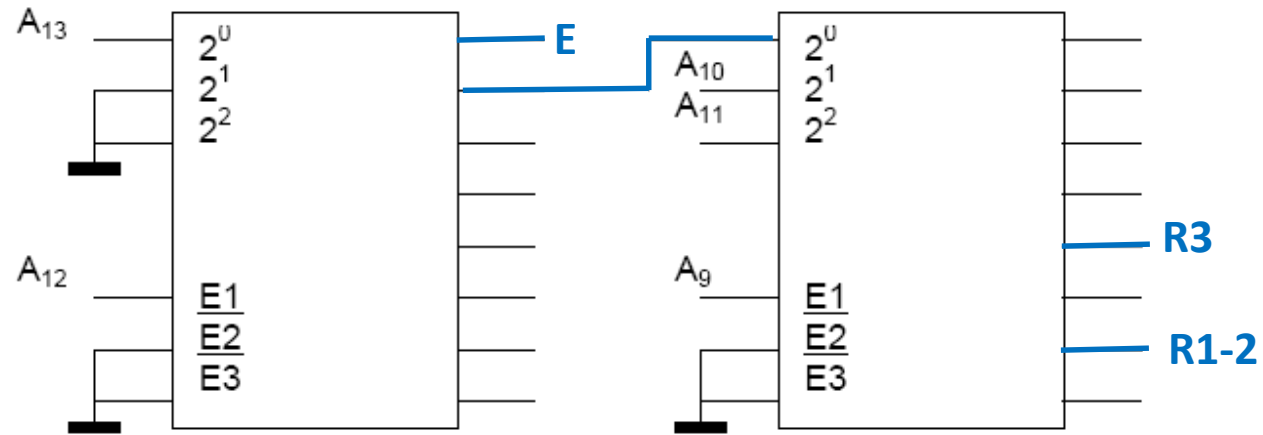




Problema 1.5

Dado el siguiente circuito encuentre el mapeo correspondiente:





Memoria	Dir. Hexa.	A15 A14 A13 A12	A11 A10 A9 A8	A7 A6 A5 A4	A3 A2 A1 A0
EPROM	1000	0 0 0 1	0 0 0 0	0 0 0 0	0 0 0 0
	1FFF	0 0 0 1	1 1 1 1	1 1 1 1	1 1 1 1
RAM3	3A00	0 0 1 1	1 0 1 0	0 0 0 0	0 0 0 0
	3BFF	0 0 1 1	1 0 1 1	1 1 1 1	1 1 1 1
RAM 1-2	3E00	0 0 1 1	1 1 1 0	0 0 0 0	0 0 0 0
	3FFF	0 0 1 1	1 1 1 1	1 1 1 1	1 1 1 1

Mapeo relativo resultado del problema:

A15=0
A14=0

0000 – 0FFFH

1000 – 1FFFH

2000 – 39FFH

3A00 – 3BFFH

3C00 – 3DFFH

3E00 – 3FFFH

4000 – FFFFH



Mapecto absoluto resultado del problema:

